



COLLEGE OF ELECTRICAL AND MECHANICAL ENGINEERING NUST

MECHATRONICS DEPARTMENT

DEGREE - 45

SYNDICATE-A

MODELING AND SIMULATION

(M&S-321)

MODULE 2 REPORT

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In module 1 we have successfully established the configuration for the Lane-Keeping Assist (LKA) system. Module 2 focuses on the development of a mathematical model suitable for simulation and control implementation. The model will translate the selected components, that are, monocular camera sensor, LQR control algorithm, dynamic bicycle vehicle dynamics, kinematic vehicle dynamics and the reference vehicle (Toyota Corolla Altis X 1.6) into a mathematical framework.

Identification of Main LKA Components

The LKA system has the four main components:

- Lane Detection System (Monocular Camera)
- Control Unit (LQR Controller)
- Steering Actuator
- Vehicle Dynamics Model

The interconnection of these components creates a closed loop control system where sensor feedback continuously adjusts steering to minimize lane deviation and heading error.

Relevant Physical Parameters

The mathematical model uses some specific parameters which were either obtained from technical documentation or estimated from literature. You should find these below:

Parameter	Symbol	Value
Vehicle Mass	m	1275 kg
Yaw Inertia	I_z	2400 kg·m ²
Distance from CG to Front Axle	l_f	1.1 m
Distance from CG to Rear Axle	l_r	1.6 m
Total Wheelbase	L	2.7 m
Front Cornering Stiffness	C_f	88,000 N/rad
Rear Cornering Stiffness	C_r	94,000 N/rad
Longitudinal Velocity	v_x	20 m/s

These parameters were selected to represent realistic highway conditions.

Dynamic Bicycle Model

1. Steering Control Equations

Lateral Dynamics and Tire Slip Modeling

Four state variables will represent the lateral and angular motion as per the dynamics bicycle model:

$$\mathbf{x} = \begin{bmatrix} y' \\ v' \\ e_y \\ v_e \end{bmatrix}$$

where:

- y' = lateral velocity of vehicle center of gravity (m/s)
- v' = yaw rate (rate of rotation about vertical axis) (rad/s)
- e_y = lateral position error (distance of CG from lane center) (m)
- v_e = heading error (angle between lane tangent and vehicle heading) (rad)

The tire slip angles are critical to capturing realistic vehicle dynamics:

$$\alpha_f = \sigma - (\theta_{v,f}) = \sigma - \left(\frac{y' - l_f v'}{v_x} \right)$$
$$\alpha_r = -(\theta_{v,r}) = - \left(\frac{y' - l_r v'}{v_x} \right)$$

where σ is the front wheel steering angle (control input) and $\theta_{v,f}$, $\theta_{v,r}$ are velocity angles at front and rear axles.

Lateral and Yaw Force Equations

Tire lateral forces are modeled linearly as proportional to slip angles through cornering stiffness coefficients:

$$F_{y,f} = C_f \alpha_f$$

$$F_{y,r} = C_r \alpha_r$$

Equations of Motion

The vehicle's lateral acceleration equation is derived from Newton's second law:

$$m(y'' + v_x v') = F_{y,f} + F_{y,r}$$

The yaw moment equation is obtained from rotational dynamics:

$$I_z v'' = l_f F_{y,f} - l_r F_{y,r}$$

Lane Error and Heading Error Dynamics

The lane relative states evolve according to:

$$e'_y = y' + v_x v_e$$

$$v'_e = v' - v'_{des}$$

where $v'_{des} = \frac{v_x}{R}$ is the required yaw rate to follow the desired path with radius of curvature R .

2. State Space Formulation

In order to obtain a Linear Time Invariant (LTI) system suitable for LQR design, the nonlinear control equations are linearized about nominal trajectory.

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}u$$

where $u = \sigma$ is the steering angle input.

The system matrices are obtained from the linearization of the equations of motion:

$$\mathbf{A} = \begin{bmatrix} -\frac{C_f + C_r}{mv_x} & \frac{l_r C_r - l_f C_f}{mv_x} - v_x & 0 & 0 \\ \frac{l_r C_r - l_f C_f}{I_z v_x} & -\frac{l_r^2 C_r + l_f^2 C_f}{I_z v_x} & 0 & 0 \\ 1 & 0 & 0 & v_x \\ 0 & 1 & 0 & 0 \end{bmatrix}$$

$$\mathbf{B} = \begin{bmatrix} \frac{C_f}{m} \\ \frac{l_f C_f}{I_z} \\ 0 \\ 0 \end{bmatrix}$$

The outputs are two lateral states that must be regulated.

$$\mathbf{y} = \mathbf{C}\mathbf{x} + \mathbf{D}u$$

$$\mathbf{C} = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \mathbf{D} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

3. LQR Controller Design

LQR minimizes the quadratic cost function:

$$J = \int_0^{\infty} (\mathbf{x}^T \mathbf{Q} \mathbf{x} + u^T \mathbf{R} u) dt$$
$$\mathbf{Q} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 100 & 0 \\ 0 & 0 & 0 & 10 \end{bmatrix}, \quad \mathbf{R} = [1]$$

The Q matrix weights penalize lateral error (e_y) most heavily (weight = 100), followed by heading error (v_e , weight = 10). The R matrix (weight = 1) constrains excessive steering input, ensuring physically realizable control commands.

The optimal feedback gain is obtained by solving the Algebraic Riccati Equation (ARE). The solution yields the Riccati matrix:

$$\mathbf{P} = \begin{bmatrix} 0.0811 & -0.1203 & 0.4896 & 0.7047 \\ -0.1203 & 0.2274 & -0.5899 & -0.6312 \\ 0.4896 & -0.5899 & 11.591 & 13.185 \\ 0.7047 & -0.6312 & 13.185 & 37.988 \end{bmatrix}$$

The optimal state feedback control law is:

$$u = -\mathbf{K}\mathbf{x}$$

where the feedback gain matrix is computed as:

$$\mathbf{K} = \mathbf{R}^{-1} \mathbf{B}^T \mathbf{P} = \begin{bmatrix} 0.7438 \\ 0.8688 \\ 10.000 \\ 23.182 \end{bmatrix}$$

- Lateral velocity gain: 0.7438 rad/m·s
- Yaw rate gain: 0.8688 rad/(rad/s)
- Lane error gain: 10.000 rad/m (primary lane correction)
- Heading error gain: 23.182 rad/rad (strong heading correction)

4. External Factors and Initial Conditions

Real world LKA systems operate in the presence of disturbances and sensor noise. The augmented state-space model incorporates these effects:

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}u + \mathbf{B}_d d + \mathbf{B}_c w_c$$

$$\mathbf{y} = \mathbf{C}\mathbf{x} + \eta$$

where:

- d represents external disturbances (crosswind effects, road banking)
- w_c represents measurement noise from the camera sensor
- η represents output measurement noise

The disturbance and noise signals are modeled as:

$$\mathbf{B}_d = \begin{bmatrix} \frac{1}{m} \\ 0 \\ 0 \\ 0 \end{bmatrix}, \mathbf{B}_c = \begin{bmatrix} 0 \\ 0 \\ 0 \\ -1 \end{bmatrix}$$

The measurement noise covariance matrix characterizes sensor uncertainty:

$$\Sigma = \begin{bmatrix} 0.0004 & 0 \\ 0 & 0.000025 \end{bmatrix}$$

These values represent:

- Lane position measurement noise (standard deviation ≈ 0.02 m)
- Heading angle measurement noise (standard deviation ≈ 0.005 rad)

The initial conditions are represented as:

$$\mathbf{x}_0 = \begin{bmatrix} 0 \\ 0 \\ 0.3 \\ 0.05 \end{bmatrix}$$

- $y'_0 = 0$ m/s: Vehicle initially moving perpendicular to lane (no lateral velocity bias)
- $v'_0 = 0$ rad/s: Zero initial yaw rate
- $e_{y,0} = 0.3$ m: Vehicle positioned 0.3 m from lane center (significant lateral error)
- $v_{e,0} = 0.05$ rad: Vehicle heading misaligned by approximately 2.9° relative to lane

Kinematic Bicycle Model

Prior we have established the mathematics for the dynamic bicycle model, now we will establish the mathematical model for the kinematic bicycle model.

There are some assumptions we will take that differ from the dynamic bicycle model.

- Zero Tire Slip Angles
- As before ideal conditions are considered

1. Equations of Motion

The vehicle pose in the global frame is described by (X, Y, Ψ) . The front steering angle will be sole input. The equations of motion are:

$$\dot{X} = v_x \cos(\psi)$$

$$\dot{Y} = v_x \sin(\psi)$$

$$\dot{\psi} = \left(\frac{v_x}{L}\right) \sigma$$

Unlike the dynamic model, the kinematic model does not define slip angles. Therefore:

$$\alpha_f = 0$$

$$\alpha_r = 0$$

2. Lane Error State

The global coordinates are transformed to road relative error states. The obtained states after linearizing and small angle approximation:

$$e_{\dot{y}} = v_x e_{\psi}$$

$$e_{\psi} = \left(\frac{v_x}{L}\right) \sigma - \frac{v_x}{R}$$

The term $\frac{v_x}{R}$ is the desired yaw rate to follow the road curvature.

3. State Space Representation

The two error-state equations are written in standard state-space form:

$$\dot{x} = Ax + Bu + B_c w_c$$

$$x = \begin{bmatrix} e_y \\ e_{\psi} \end{bmatrix}$$

$$A = \begin{bmatrix} 0 & v_x \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 20 \\ 0 & 0 \end{bmatrix}$$

$$B = \begin{bmatrix} 0 \\ \frac{v_x}{l} \end{bmatrix} = \begin{bmatrix} 0 \\ 7.407 \end{bmatrix}$$

$$B_c = [0]$$

$$C = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad D = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

4. LQR Controller Design

The LQR controller minimizes the quadratic cost function:

$$J = \int_0^{\infty} (\mathbf{x}^T \mathbf{Q} \mathbf{x} + u^T \mathbf{R} u) dt$$

$$Q = \begin{bmatrix} 100 & 0 \\ 0 & 10 \end{bmatrix} \quad R = [1]$$

Solving the ARE we get:

$$P = \begin{bmatrix} 13.693 & 1.852 \\ 1.852 & 1.635 \end{bmatrix}$$

$$K = [10 \quad 8]$$

The optimal control is therefore:

$$\sigma^* = -K x = -13.693 e_y - 12.114 e_\psi$$

5. Initial Conditions

The initial conditions are similar to that of the dynamic bicycle model:

$$x_0 = \begin{bmatrix} 0.3 \text{ m} \\ 0.05 \text{ rad} \end{bmatrix}$$

Conclusion

In this module we have formulated the mathematics necessary for the simulation of a LKA system for both the kinematic bicycle model and the dynamic bicycle model. Thus, the mathematical development for module 2 is now complete and ready for MATLAB/Simulink simulation in module 3.